

It is nothing short of amazing and humbling fact that even after 120 years of trying to grow microbes in the laboratory, we have succeeded in culturing only 0.1% of the microorganisms around us.

iv) Differential medium

Differential medium distinguishes between different groups of bacteria and permit tentative identification of microorganisms based on their biological charaterstictics as they cause a visible change in the medium. We can differentiate haemolytic and non-haemolytic patterns of bacteria using blood agar. Differential medium is otherwise called indicator medium as it distinguishes one organism from another growing on the same plate by the formation of pigments due to its biochemical and physiological nature. Example: MacConkey agar medium has neutral red dye. Lactose fermentors form pink coloured colonies and non fermentors form colourless translucent colonies on it (Figure 5.7).



Figure 5.7: Growth of microorganisms on MacConkey agar (Lactose fermenting bacterial colonies appears pink)

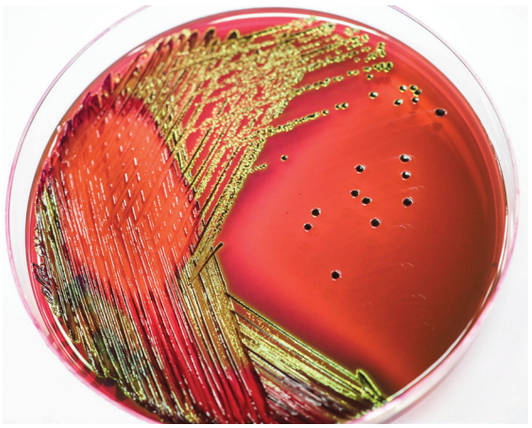
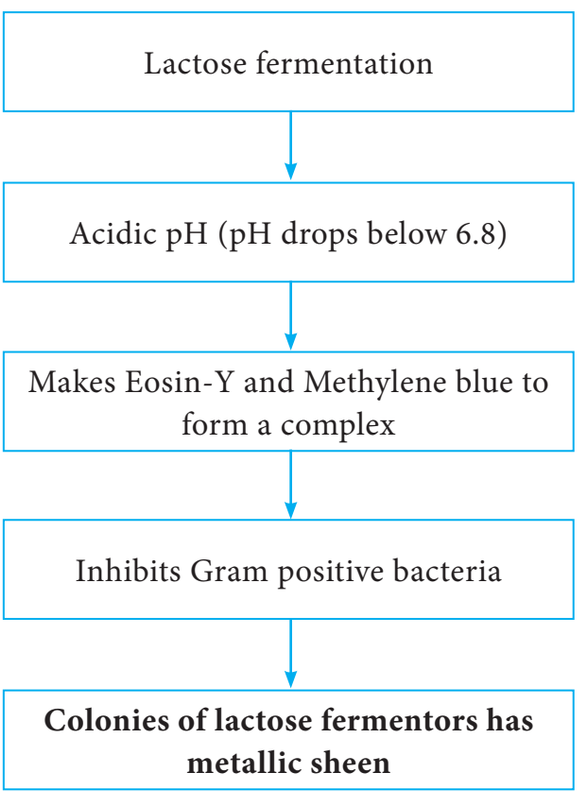


Figure 5.8: Growth of lactose fermenting bacteria on EMB Medium

Eosin Methylene Blue (EMB) agar medium is also a differential medium. It is used to differentiate lactose fermentors from non-lactose fermentors. It has lactose sugar and two dyes namely Eosin –Y and Methylene blue. These dyes act as inhibitory agent towards Gram positive bacteria. Example: Lactose fermentors such as faecal *Escherichia coli* show metallic sheen and non lactose fermentors such as *Enterococcus* do not show metallic sheen. (Figure 5.8).



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